Cryptography

ElGamal Public Key Encryption

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ElGamal Public Key Encryption

Public-Key Cryptography



UC 2.

Public-Key Cryptography and Key Establishment

- Introduction to Public-Key Cryptography
- Number Theory and Modular Inverses
- RSA Cryptosystem
- RSA Implementation and Security
- Diffie–Hellman Key Exchange and the Discrete Log Problem
- Key Establishment and Security Attacks

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Bruce Schneier



Twofish,
Blowfish

Vincent Rijmen
and Joan Daemen



AES

Modern private
key encryption

Rivest, Shamir
& Aldeman



Public-k
encrypt

Ron
Rivest



Hashing

Whitfield
Diffie



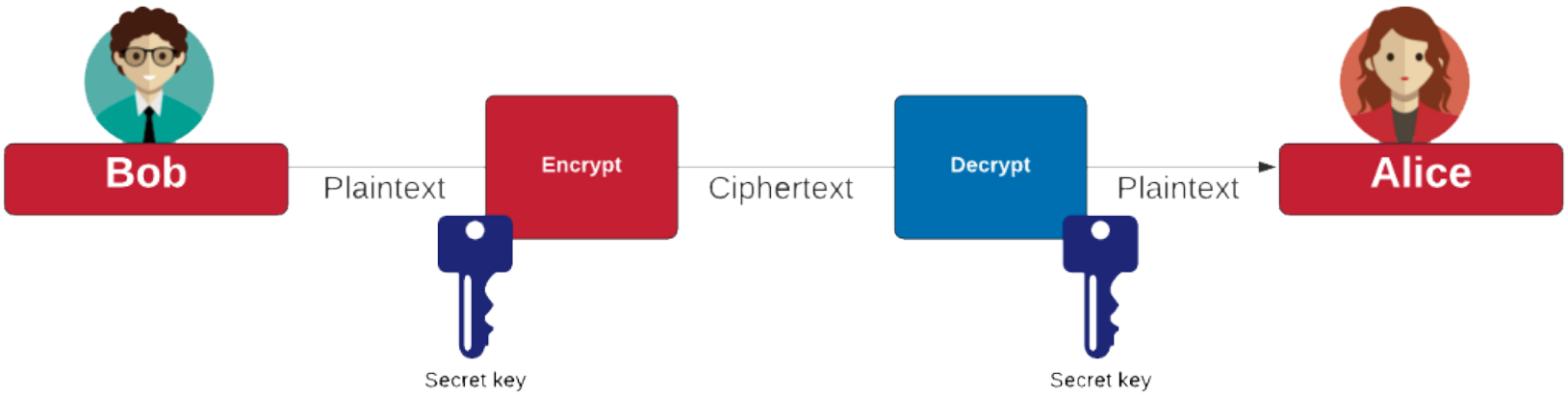
Key
interchange

Phil
Zimmerman



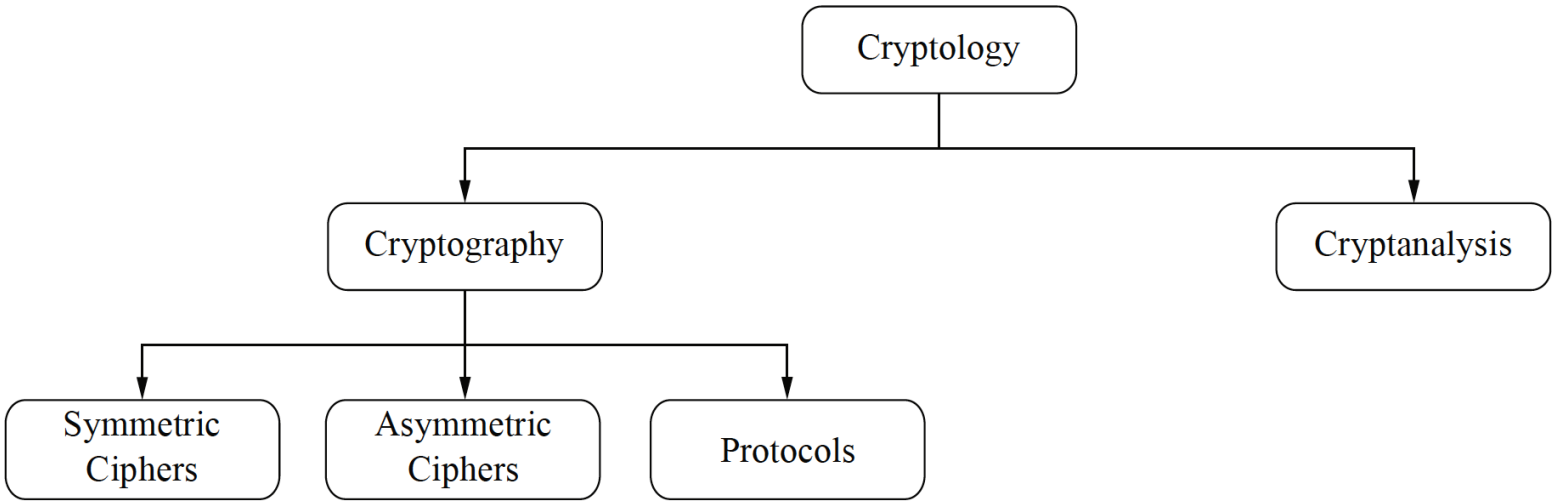
PGP
Encryption

Limitations of Symmetric Cryptography



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Introduction to Public-Key Cryptography



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Evolution of Public-Key Cryptography

1976: New concept
Public-key cryptography and secure key establishment

Public-key encryption and digital signatures become practical security tools



Main families:
RSA $\longrightarrow$ integer factorization
ElGamal / DH $\longrightarrow$ discrete logarithm
ECC $\longrightarrow$ elliptic-curve discrete logarithm

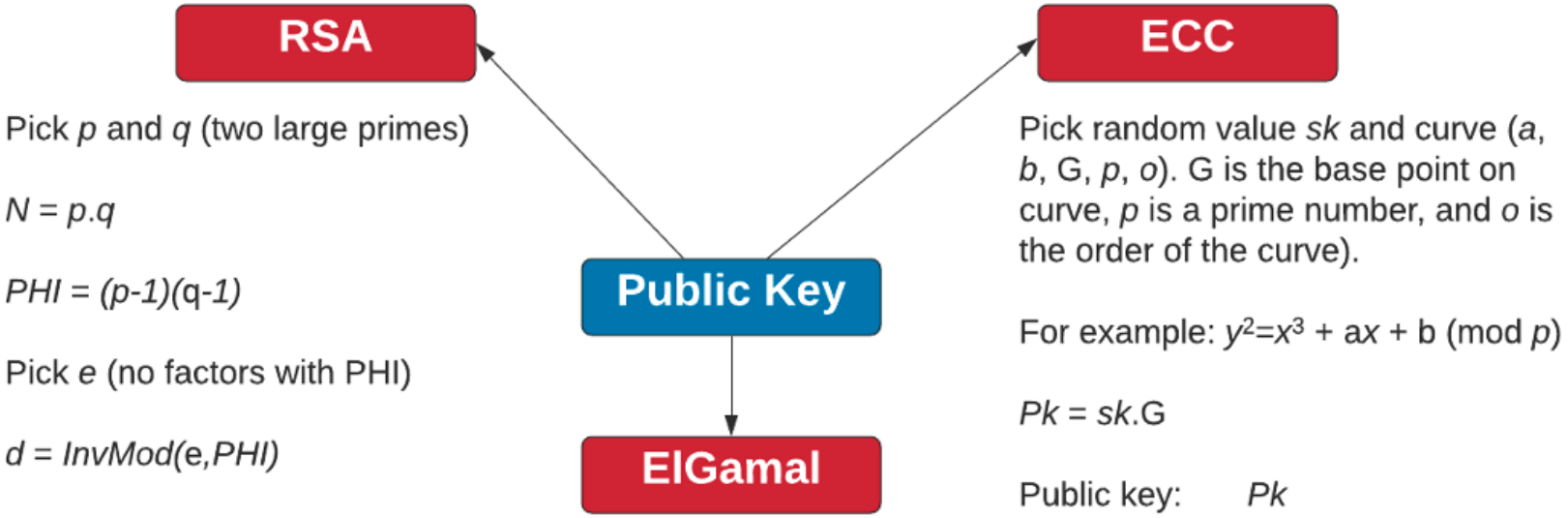
Historical Impact

Public-key cryptography enabled **key establishment, encryption, and digital signatures** without requiring a previously shared secret key. Public keys still need an authentication mechanism to prevent impersonation.

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Elgamal Cryptography Algorithm

Public Key Encryption Methods



Pick p and q (two large primes)

$N = p.q$

$PHI = (p-1)(q-1)$

Pick e (no factors with PHI)

$d = InvMod(e,PHI)$

Public: (e,N)
Private: (d,N)

$C = M^e \pmod N$
$P = C^d \pmod N$

Pick random value x, generator (g) and prime (p)

$Y = g^x \pmod p$

Public key: (Y,p)
Private: (x,p)

Pick random value sk and curve (a, b, G, p, o). G is the base point on curve, p is a prime number, and o is the order of the curve).

For example: $y^2=x^3 + ax + b \pmod p$

$Pk = sk.G$

Public key: Pk
Private key: sk

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Public-Key Cryptography

ElGamal Encryption Scheme

- **ElGamal** is a public-key encryption scheme proposed by Taher ElGamal in 1985.
- Its security is based on the difficulty of the Discrete Logarithm Problem (DLP).
- We study ElGamal over the multiplicative group $\mathbb{Z}_p^*$, where p is prime.
- The scheme uses a private key x and a public key (p, α, β), where $\beta = \alpha^x \bmod p$.

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Elgamal Cryptography Algorithm

From Diffie–Hellman Key Exchange to Elgamal Encryption

- We consider two parties, Alice and Bob. If Alice wants to send an encrypted message x to Bob, both parties first perform a Diffie-Hellman key exchange to derive a shared key k_M.
- A large prime p and a primitive element α have been generated.
- Alice uses this key as a multiplicative mask to encrypt x as $y \equiv x \cdot k_M \bmod p$.

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ElGamal Encryption

From Diffie–Hellman to ElGamal Encryption

- Alice wants to send a message m securely to Bob.
- Bob has a public key based on a large prime p, a generator α, and a public value derived from his private key.
- Alice uses an ephemeral Diffie–Hellman value to derive a shared secret k_M with Bob.
- The shared secret is used as a multiplicative mask:

$$c_2 = m \cdot k_M \pmod{p}$$

- Alice also sends the ephemeral public value c_1, which allows Bob to reconstruct the same shared secret and recover m.

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Principle of Elgamal Encryption





(a) choose $d = k_{pr,B} \in \{2, \ldots, p-2\}$
(b) compute $\beta = k_{pub,B} \equiv \alpha^d \mod p$

β ←

(c) choose $i = k_{pr,A} \in \{2, \ldots, p-2\}$
(d) compute $k_E = k_{pub,A} \equiv \alpha^i \mod p$

k_E →

(f) compute $k_M \equiv k_E^d \mod p$

(e) compute $k_M \equiv \beta^i \mod p$
(g) encrypt message $x \in \mathbb{Z}_p^*$
$y \equiv x \cdot k_M \mod p$

y →

(h) decrypt $x \equiv y \cdot k_M^{-1} \mod p$

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ElGamal Encryption

Main Phases of the ElGamal Cryptosystem

1. **Key Generation**
   - Generate the public and private keys.
2. **Encryption**
   - Use the receiver's public key to encrypt the message.
3. **Decryption**
   - Use the receiver's private key to recover the message.

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Principle of Elgamal Encryption



Alice



Bob

choose large prime p
choose primitive element $\alpha \in \mathbb{Z}_p^*$
or in a subgroup of $\mathbb{Z}_p^*$
choose $k_{pr} = d \in \{2, \ldots, p-2\}$
compute $k_{pub} = \beta = \alpha^d \mod p$

$k_{pub} = (p, \alpha, \beta)$ ←

choose $i \in \{2, \ldots, p-2\}$
compute ephemeral key
$k_E \equiv \alpha^i \mod p$
compute masking key
$k_M \equiv \beta^i \mod p$
encrypt message $x \in \mathbb{Z}_p^*$
$y \equiv x \cdot k_M \mod p$

→ (k_E, y)

compute masking key
$k_M \equiv k_E^d \mod p$
decrypt $x \equiv y \cdot k_M^{-1} \mod p$

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Principle of Elgamal Encryption





message $x = 26$

generate $p = 29$ and $\alpha = 2$
choose $k_{pr,B} = d = 12$
compute $\beta = \alpha^d \equiv 7 \bmod 29$

$k_{pub,B}=(p,\alpha,\beta)$ ←

choose $i = 5$
compute $k_E = \alpha^i \equiv 3 \bmod 29$
compute $k_M = \beta^i \equiv 16 \bmod 29$
encrypt $y = x \cdot k_M \equiv 10 \bmod 29$

y,k_E →

compute $k_M = k_E^d \equiv 16 \bmod 29$
decrypt
$x = y \cdot k_M^{-1} \equiv 10 \cdot 20 \equiv 26 \bmod 29$

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ElGamal

ElGamal - Key Generation

Bob chooses:

- a large prime p
- a generator α
- a private key $d = k_{pr,B}$

Bob computes
$\beta = k_{pub,B} \equiv \alpha^d \pmod{p}$.

$$\text{Public key} = \{p, \alpha, \beta\}$$

$$\text{Private key} = \{d\}$$

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ElGamal - Encryption

Alice receives Bob's public value β.

Alice chooses a random private value

$$i = k_{pr,A}.$$

Then she computes

$$k_E = k_{pub,A} \equiv \alpha^i \pmod{p}.$$

Using Bob's public value, Alice computes the shared key

$$k_M \equiv \beta^i \pmod{p}.$$

For a message $x \in \mathbb{Z}_p^*$, Alice computes

$$y \equiv x \cdot k_M \pmod{p}.$$

$$\text{Ciphertext} = \{k_E, y\}$$

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ElGamal - Decryption

Bob receives the ciphertext $\{k_E, y\}$.

Using his private key d, Bob computes the shared key

$$k_M \equiv k_E^d \pmod{p}.$$

Since

$$k_E = \alpha^i,$$

then

$$k_M \equiv (\alpha^i)^d \equiv \alpha^{id} \pmod{p}.$$

Alice computed the same value:

$$k_M \equiv \beta^i \equiv (\alpha^d)^i \equiv \alpha^{di} \pmod{p}.$$

Bob recovers the message using

$$x \equiv y \cdot k_M^{-1} \pmod{p}.$$

Recovered message $= \{x\}$

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ElGamal - Example

Let $p = 23$ and $\alpha = 5$.

Bob: Key Generation
Bob chooses the private key $d = 6$ and computes
$\beta = \alpha^d \bmod p = 5^6 \bmod 23 = 8$.
Thus,
Public key $= \{23, 5, 8\}$
Private key $= \{6\}$

Alice: Encryption
Let the message be $x = 10$.
Alice chooses $i = 3$ and computes
$k_E = \alpha^i \bmod p = 5^3 \bmod 23 = 10$,
$k_M = \beta^i \bmod p = 8^3 \bmod 23 = 6$,
$y = x \cdot k_M \bmod p = 10 \cdot 6 \bmod 23 = 14$.

$$\text{Ciphertext} = \{k_E, y\} = \{10, 14\}$$

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ElGamal - Example: Decryption

Bob receives the ciphertext

$$\{k_E, y\} = \{10, 14\}.$$

Using his private key $d = 6$, Bob computes

$$k_M \equiv k_E^d \pmod{p}.$$

Thus,

$$k_M \equiv 10^6 \pmod{23} = 6.$$

Now Bob computes the inverse

$$k_M^{-1} \equiv 6^{-1} \pmod{23} = 4.$$

Finally,

$$\begin{aligned} x &\equiv y \cdot k_M^{-1} \pmod{p}, \\ x &\equiv 14 \cdot 4 \pmod{23}, \\ x &\equiv 56 \pmod{23}, \\ \boxed{x &= 10}. \end{aligned}$$

Recovered message $= \{10\}$

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Security of ElGamal

The security of ElGamal is based on the difficulty of the **Discrete Logarithm Problem**.

From the public information
$\beta \equiv \alpha^d \pmod{p}$,
an attacker knows α, β, and p.
However, recovering the private key d requires solving
$\alpha^d \equiv \beta \pmod{p}$.

For properly chosen parameters, this problem is computationally hard.

The random value i used during encryption must also be:

- random,
- secret,
- different for each encryption.

Reusing i can reveal information about the encrypted messages.

Security depends on strong parameters and fresh randomness.

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Exercise 1: ElGamal Cryptosystem

Let $p = 47$, $\alpha = 5$. Bob chooses the private key $d = 9$. Alice wants to send the message $x = 17$ and chooses the random value $i = 13$.

Tasks

1. Compute Bob's public value.
2. Compute Alice's ephemeral public value.
3. Compute the shared value.
4. Encrypt the message.
5. Write the ciphertext.
6. Decrypt the ciphertext using Bob's private key.
7. Verify that the recovered message is $x = 17$.

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Exercise 2: Reusing the Random Value

An ElGamal system uses $p = 59$, $\alpha = 2$. Bob's private key is $d = 17$, so his public value is $\beta = 33$.
Alice encrypts two different messages: $x_1 = 12$ and $x_2 = 25$. However, she makes a serious mistake: she uses the same random value $i = 9$ for both messages.

Tasks

1. Compute both ciphertext values $y_1 = x_1 k_M \bmod p$ and $y_2 = x_2 k_M \bmod p$.
2. Compare the two ciphertexts $\{k_E, y_1\}$ and $\{k_E, y_2\}$.
3. Suppose an attacker knows x_1 and its ciphertext. Show how the attacker can recover k_M.
4. Use this information to recover the unknown message x_2.
5. Explain why the random value i must never be reused.

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Solution: Reusing the Random Value

Alice uses the same random value $i = 9$.

$k_E = \alpha^i \bmod p = 2^9 \bmod 59 = \boxed{40}$

$k_M = \beta^i \bmod p = 33^9 \bmod 59 = \boxed{39}$

The ciphertexts are

$y_1 = 12(39) \bmod 59 = \boxed{55}$

$y_2 = 25(39) \bmod 59 = \boxed{31}$

Thus,

$\{k_E, y_1\} = \{40, 55\}$

$\{k_E, y_2\} = \{40, 31\}$

If the attacker knows $x_1 = 12$:

$12^{-1} \equiv 5 \pmod{59}$

$k_M \equiv 55(5) \bmod 59 = \boxed{39}$

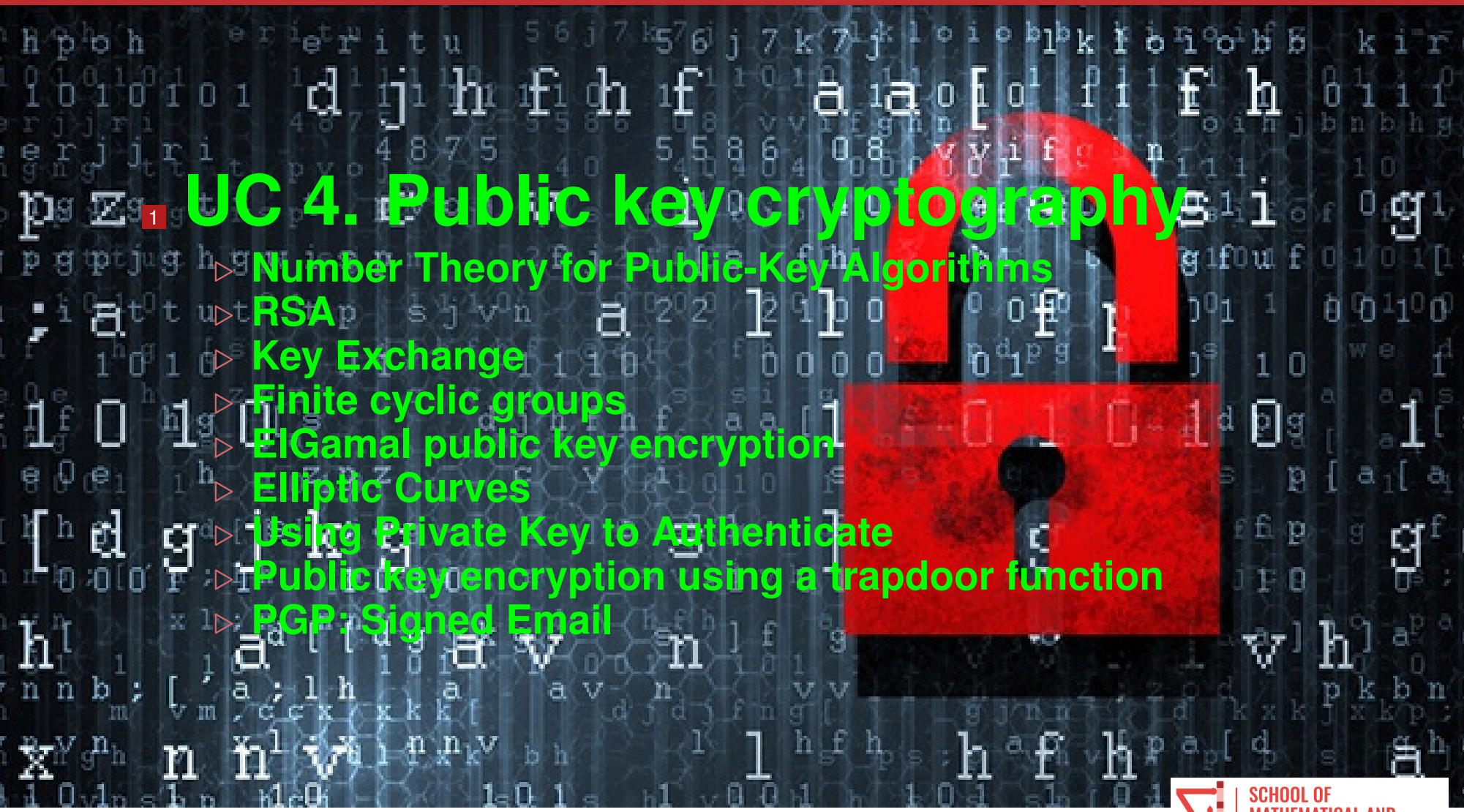
Since $39^{-1} \equiv 56 \pmod{59}$,

$x_2 \equiv 31(56) \bmod 59 = \boxed{25}$.

Conclusion: Reusing i allows an attacker to recover other messages.

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Cryptography Fundamentals



UC 4. Public key cryptography

- Number Theory for Public-Key Algorithms
- RSA
- Key Exchange
- Finite cyclic groups
- ElGamal public key encryption
- Elliptic Curves
- Using Private Key to Authenticate
- Public key encryption using a trapdoor function
- PGP: Signed Email

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